

TidyTuesday Week 41

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Thursday 8th October, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we grab two economic time series that have never met, never will meet, and have no business being in the same spreadsheet, and then we ask them to explain each other. As always, the house rules apply: these pairs are drawn at random, no causal story is intended, and if you walk away from this convinced that natural gas prices control foreign bond holdings, that's a you problem, not a statistics problem. Spurious correlation isn't a bug here. It's the entire point.

We opened Monday with the Producer Price Index for residential natural gas in the Middle Atlantic versus the percent of syndicated loans at foreign bank branches rated "Acceptable" risk. The raw regression came out swinging with an R-squared of 0.44 and a p-value of 0.001, which in less careful hands would get a headline like "Gas Prices Secretly Rule the Banking System." And then we detrended it, and the whole thing evaporated like a politely-rated loan in a downturn: R-squared cratered to 0.02 and the p-value strolled up to 0.56. Classic trend mirage. Two series that happened to wander in roughly compatible directions, holding hands only until we took the trend away, at which point they pretended they'd never met. Beautiful. Textbook. Framed and hung in the gallery.

Tuesday served up bank equity-to-assets ratios in the South Atlantic versus the dollar-to-SDR exchange rate, and honestly this one never even made it out of the gate. Yes, the raw version was technically "significant" at $p=0.003$, but with an R-squared of 0.067 it was explaining roughly nothing with great confidence, which is the statistical equivalent of being extremely sure about a whisper. Detrended, it collapsed to an R-squared of 0.004 and a p-value of 0.46, so it joins Monday in the trend-mirage bin. Two for two on relationships that only exist if you squint at the long-run drift and believe really hard.

Then Wednesday actually got interesting, and I'm as uncomfortable about it as you are. Hong Kong's holdings of U.S. long-term corporate bonds against commercial carbon dioxide emissions from natural gas pipelines. The raw regression showed a positive relationship (R-squared 0.24), and here's the plot twist: when we detrended, it didn't just survive, it got slightly stronger (R-squared 0.25, $p=0.003$) and flipped sign to negative. That's the rare bird this project pretends to be hunting for: a relationship that holds up after we strip the trends out, which makes it harder to wave away as mere shared drift. Do I have a theory? Sure, in the sense that I can string words together: maybe years of heavier pipeline gas use loosely coincide with periods when Hong Kong investors felt differently about U.S. corporate paper. Is that theory worth the electrons it's printed on? Absolutely not. It's a tiny annual sample, two discontinued-ish datasets duct-taped together,

and I would not bet a sandwich on it. But as far as this gloriously pointless exercise goes, it's the week's one genuine "huh." Tune in next week, when we will once again torture unrelated numbers until they confess to things they didn't do.

2 Date: 2026-10-06

Series ID: PCU2212102212101122

This series is titled Producer Price Index by Industry: Natural Gas Distribution: Residential Natural Gas for Middle Atlantic Census Division and has a frequency of Monthly. The units are Index Jun 2012=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2012-06-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: EFAOXFBNQ

This series is titled Percent of Value of Loans Made Under Participation or Syndication for All Commercial and Industry Loans, Other Risk (Acceptable), U.S. Branches and Agencies of Foreign Banks (DISCONTINUED) and has a frequency of Quarterly, 2nd Month's 1st Full Week. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2012-07-01 and the observation end date is 2017-04-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_EFAOXFBNQ	R-squared:	0.444
Model:	OLS	Adj. R-squared:	0.413
Method:	Least Squares	F-statistic:	14.37
Date:	Tue, 06 Oct 2026	Prob (F-statistic):	0.00134
Time:	17:29:15	Log-Likelihood:	-69.835
No. Observations:	20	AIC:	143.7
Df Residuals:	18	BIC:	145.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	104.3514	21.206	4.921	0.000	59.798	148.904
value_fred_PCU2212102212101122	-0.8266	0.218	-3.790	0.001	-1.285	-0.368

Omnibus:	2.921	Durbin-Watson:	1.600
Prob(Omnibus):	0.232	Jarque-Bera (JB):	1.265
Skew:	0.146	Prob(JB):	0.531
Kurtosis:	1.803	Cond. No.	1.10e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

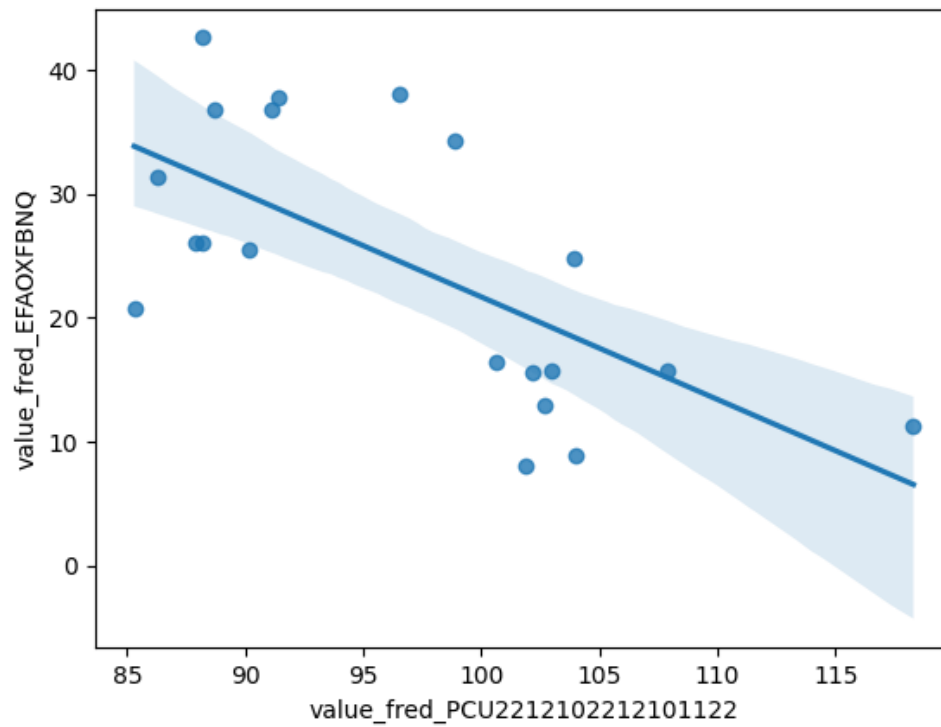


Figure 1: Raw Regression Plot for 2026-10-06

[2] The condition number is large, $1.1\text{e}+03$. This might indicate that there are strong multicollinearity or other numerical problems.

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_EFAOXFBNQ_detrended	R-squared:	0.019
Model:	OLS	Adj. R-squared:	-0.035
Method:	Least Squares	F-statistic:	0.3538
Date:	Tue, 06 Oct 2026	Prob (F-statistic):	0.559
Time:	17:29:15	Log-Likelihood:	-69.114
No. Observations:	20	AIC:	142.2
Df Residuals:	18	BIC:	144.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-4.379e-12	1.807	-2.42e-12	1.000	-3.796	3.796
value_fred_PCU2212102212101122_detrended	-0.2983	0.502	-0.595	0.559	-1.352	0.755

Omnibus:	0.524	Durbin-Watson:	1.639
Prob(Omnibus):	0.770	Jarque-Bera (JB):	0.580
Skew:	0.060	Prob(JB):	0.748
Kurtosis:	2.175	Cond. No.	3.60

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

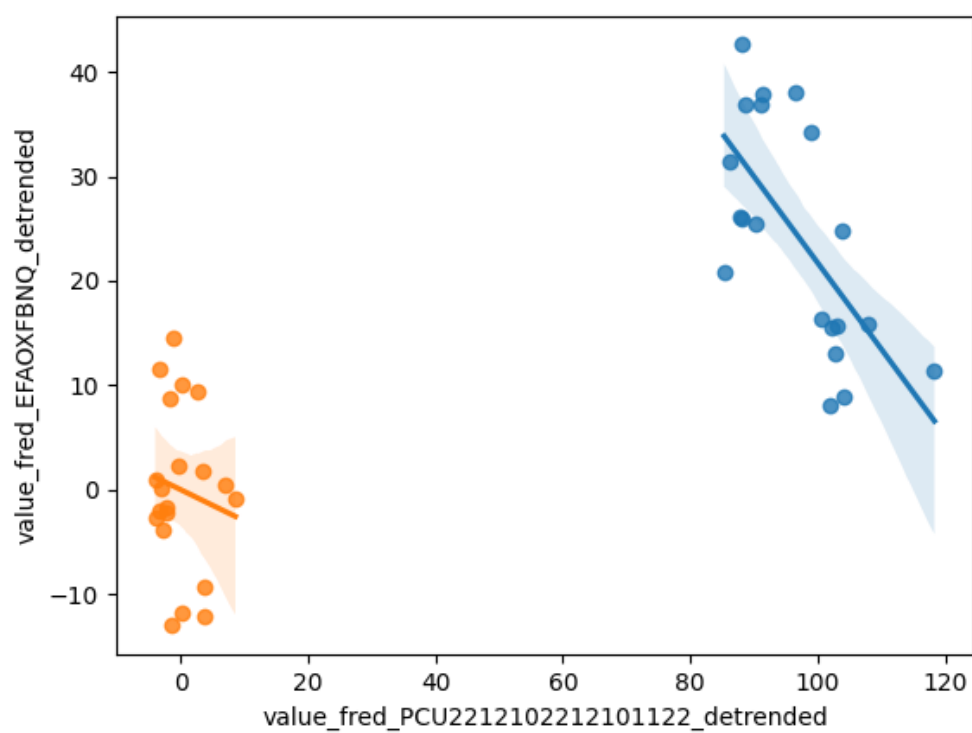


Figure 2: Detrended Regression Plot for 2026-10-06

3 Date: 2026-10-07

Series ID: EQTA35

This series is titled Total Equity to Total Assets, Banks with Total Assets from \$1B to \$10B, South Atlantic Census Division (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1988-01-01 and the observation end date is 2020-07-01. The popularity of this series is 1.

Series ID: CCUSSP01IFM650N

This series is titled Currency Conversions: US Dollar Exchange Rate: Spot, End of Period: USD: National Currency for SDR and has a frequency of Monthly. The units are Special Drawing Rights and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1975-01-01 and the observation end date is 2023-12-01. The popularity of this series is 13.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_CCUSSP01IFM650N	R-squared:	0.067
Model:	OLS	Adj. R-squared:	0.060
Method:	Least Squares	F-statistic:	9.266
Date:	Wed, 07 Oct 2026	Prob (F-statistic):	0.00283
Time:	18:02:18	Log-Likelihood:	133.95
No. Observations:	131	AIC:	-263.9
Df Residuals:	129	BIC:	-258.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.2811	0.049	26.190	0.000	1.184	1.378
value_fred_EQTA35	0.0142	0.005	3.044	0.003	0.005	0.023

Omnibus:	3.573	Durbin-Watson:	0.175
Prob(Omnibus):	0.168	Jarque-Bera (JB):	2.476
Skew:	0.161	Prob(JB):	0.290
Kurtosis:	2.409	Cond. No.	67.4

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

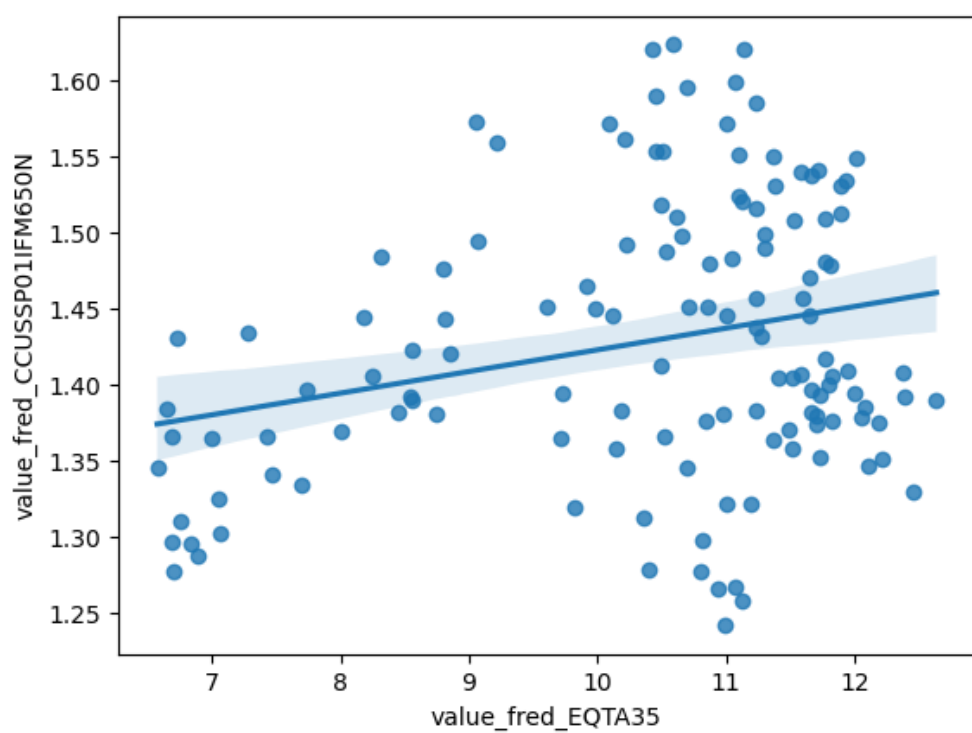


Figure 3: Raw Regression Plot for 2026-10-07

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_CCUSP01IFM650N_detrended	R-squared:	0.004
Model:	OLS	Adj. R-squared:	-0.003
Method:	Least Squares	F-statistic:	0.5577
Date:	Wed, 07 Oct 2026	Prob (F-statistic):	0.457
Time:	18:02:18	Log-Likelihood:	139.59
No. Observations:	131	AIC:	-275.2
Df Residuals:	129	BIC:	-269.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t 	[0.025	0.975]
const	3.906e-14	0.007	5.32e-12	1.000	-0.015	0.015
value_fred_EQTA35_detrended	-0.0055	0.007	-0.747	0.457	-0.020	0.009

Omnibus:	10.155	Durbin-Watson:	0.184
Prob(Omnibus):	0.006	Jarque-Bera (JB):	4.169
Skew:	0.123	Prob(JB):	0.124
Kurtosis:	2.162	Cond. No.	1.00

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

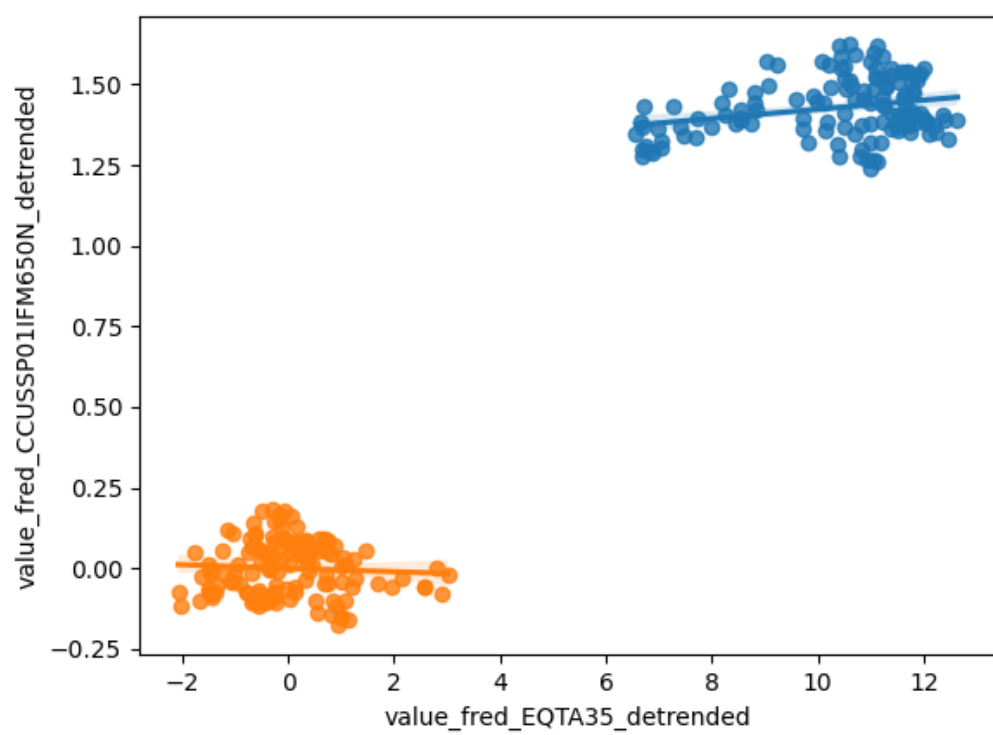


Figure 4: Detrended Regression Plot for 2026-10-07

4 Date: 2026-10-08

Series ID: FORLTCORPPOS42005

This series is titled Foreign Portfolio Holdings of U.S. Long-Term Corporate Bonds: Hong Kong and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1984-12-01 and the observation end date is 2026-07-01. The popularity of this series is 2.

Series ID: EMISSCO2VNGCCBA

This series is titled Commercial Carbon Dioxide Emissions, Natural Gas (Pipeline) for United States (DISCONTINUED) and has a frequency of Annual. The units are Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1980-01-01 and the observation end date is 2017-01-01. The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_EMISSCO2VNGCCBA	R-squared:	0.242
Model:	OLS	Adj. R-squared:	0.217
Method:	Least Squares	F-statistic:	9.875
Date:	Thu, 08 Oct 2026	Prob (F-statistic):	0.00367
Time:	18:03:19	Log-Likelihood:	-586.01
No. Observations:	33	AIC:	1176.
Df Residuals:	31	BIC:	1179.
Df Model:	1		
Covariance Type:	nonrobust		

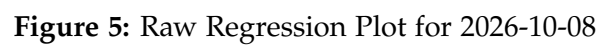
	coef	std err	t	P> t	[0.025	0.975]
const	1.558e+08	3.05e+06	51.049	0.000	1.5e+08	1.62e+08
value_fred_FORLTCORPPOS42005	770.8558	245.299	3.143	0.004	270.565	1271.147

Omnibus:	0.800	Durbin-Watson:	0.469
Prob(Omnibus):	0.670	Jarque-Bera (JB):	0.606
Skew:	-0.321	Prob(JB):	0.739
Kurtosis:	2.832	Cond. No.	1.69e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.69e+04. This might indicate that there are strong multicollinearity or other numerical problems.



4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_EMISSCO2VNGCCBA_detrended	R-squared:	0.247
Model:	OLS	Adj. R-squared:	0.223
Method:	Least Squares	F-statistic:	10.18
Date:	Thu, 08 Oct 2026	Prob (F-statistic):	0.00324
Time:	18:03:19	Log-Likelihood:	-570.19
No. Observations:	33	AIC:	1144.
Df Residuals:	31	BIC:	1147.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975
const	-1.899e-06	1.39e+06	-1.37e-12	1.000	-2.83e+06	2.83e+06
value_fred_FORLTCORPPOS42005_detrended	-895.4645	280.652	-3.191	0.003	-1467.857	-323.071

Omnibus:	5.377	Durbin-Watson:	1.047
Prob(Omnibus):	0.068	Jarque-Bera (JB):	3.796
Skew:	-0.669	Prob(JB):	0.150
Kurtosis:	3.985	Cond. No.	4.94e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.94e+03. This might indicate that there are strong multicollinearity or other numerical problems.

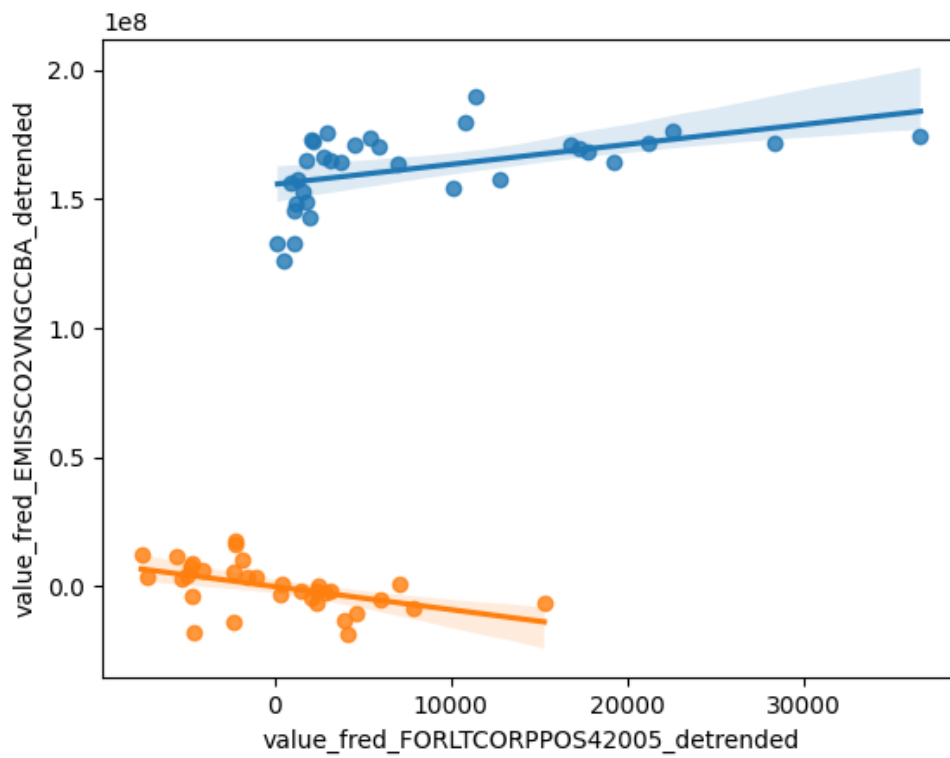


Figure 6: Detrended Regression Plot for 2026-10-08